

Abstract

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This exemplar demonstrates a **data descriptor** workflow in which the schema, file inventory, provenance chain, license boundary, and validation gate are treated as first-class research artifacts rather than afterthoughts. It ships a small, public, synthetic demonstration dataset (two CSV files under `data/fixtures/`) and a machine-readable descriptor (`data/example_descriptor.json`) that declares each file's media type, sha256 checksum, and row count alongside a six-field data dictionary with typed constraints. A tested validation library (`src/data_descriptor/`) checks the descriptor's shape, safety, and completeness; recomputes each declared checksum and row count against the bytes on disk; and emits a deterministic, metadata-only release manifest suitable for pre-publication review. Every figure and quantitative claim in this manuscript is produced by that library and regenerated on demand, so the

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prose describes structure and provenance rather than transcribing values that would drift. This is a template with a demonstration dataset: it makes no scientific claim about the data, only about how to describe and release a dataset responsibly.